

A Appendix: Proofs

A.1 Notation

We consider a classifier of the form

$$f_{\Theta}(x) = W\phi_{\Theta}(x),$$

where $W \in \mathbb{R}^{C \times m}$ is the linear classifier and $\phi_{\Theta}(x) \in \mathbb{R}^m$ is the feature representation. Let w_k denote the k -th row of W , and define the logit score for class k as

$$s_k(x) = w_k^{\top} \phi_{\Theta}(x).$$

The ℓ_{∞} adversarial ball is denoted by

$$\mathcal{B}_{\epsilon}(x) = \{x' \mid \|x' - x\|_{\infty} \leq \epsilon\}.$$

The robust risk is

$$R_{\text{robust}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{x' \in \mathcal{B}_{\epsilon}(x)} \ell(f_{\Theta}(x'), y) \right],$$

where $\ell(\cdot, \cdot)$ denotes the classification loss.

A.2 Proof of Theorem 1

Assumptions. We assume normalized features and classifier weights, i.e.,

$$\|\phi_{\Theta}(x)\|_2 = 1, \quad \|w_k\|_2 = 1 \quad \forall k,$$

which can be enforced without loss of generality. We further assume that ϕ_{Θ} is L -Lipschitz under ℓ_{∞} perturbations, as stated in the theorem.

Proof. Fix a sample (x, y) such that

$$y = \arg \max_k s_k(x).$$

Define the logit margin

$$\gamma(x) = s_y(x) - \max_{k \neq y} s_k(x).$$

Step 1: Margin lower bound from ETF geometry. Since both w_k and $\phi_{\Theta}(x)$ are unit-norm, we have

$$s_k(x) = \cos(\angle(w_k, \phi_{\Theta}(x))).$$

Let

$$k^* = \arg \max_{k \neq y} s_k(x).$$

Under the approximate ETF assumption, the angle between w_y and w_{k^*} is at least $\theta_{\min}$.

The configuration minimizing the margin occurs when $\phi_{\Theta}(x)$ lies in the two-dimensional subspace spanned by w_y and w_{k^*} . Elementary geometric arguments then yield

$$\gamma(x) \geq \sin(\theta_{\min}/2).$$

Step 2: Stability under adversarial perturbations. Let $x' = x + \delta$ with $\|\delta\|_{\infty} \leq \epsilon$. By the L -Lipschitz assumption,

$$\|\phi_{\Theta}(x') - \phi_{\Theta}(x)\|_2 \leq L\epsilon.$$

For any class k , we have

$$|s_k(x') - s_k(x)| = |w_k^{\top}(\phi_{\Theta}(x') - \phi_{\Theta}(x))| \leq L\epsilon.$$

Therefore, the margin at x' satisfies

$$\gamma(x') \geq \gamma(x) - 2L\epsilon.$$

Step 3: Robustness condition. Combining the two bounds,

$$\gamma(x') \geq \sin(\theta_{\min}/2) - 2L\epsilon.$$

Thus, if

$$\epsilon < \frac{\sin(\theta_{\min}/2)}{L},$$

the margin remains strictly positive and the predicted label is invariant to all perturbations in $\mathcal{B}_{\epsilon}(x)$. This proves the theorem. $\square$

A.3 Proof of Corollary 1

Proof. Fix a sample (x, y) and define the logit score $s_k(x) = w_k^{\top} \phi_{\Theta}(x)$. Let

$$\gamma(x) = s_y(x) - \max_{k \neq y} s_k(x)$$

denote the logit margin at x .

Consider any perturbation $x' = x + \delta$ with $\|\delta\|_{\infty} \leq r$. By the L -Lipschitz assumption on ϕ_{Θ} ,

$$\|\phi_{\Theta}(x') - \phi_{\Theta}(x)\|_2 \leq Lr.$$

Assuming $\|w_k\|_2 = 1$ for all k , we have for each class k ,

$$|s_k(x') - s_k(x)| = |w_k^{\top}(\phi_{\Theta}(x') - \phi_{\Theta}(x))| \leq Lr.$$

Let $k^* = \arg \max_{k \neq y} s_k(x)$. Then

$$s_y(x') - s_{k^*}(x') \geq (s_y(x) - Lr) - (s_{k^*}(x) + Lr) = \gamma(x) - 2Lr.$$

Therefore, if $r \leq \gamma(x)/(2L)$, the right-hand side remains non-negative, implying

$$s_y(x') \geq s_{k^*}(x') \geq s_k(x') \quad \forall k \neq y,$$

and the predicted label is unchanged within the ℓ_{∞} ball of radius r . Thus the sample-wise robust radius satisfies

$$r(x) \geq \frac{\gamma(x)}{2L}.$$

$\square$

A.4 Proof of Theorem 2

Assumptions. We assume that the per-sample loss is bounded,

$$0 \leq \ell(f_{\Theta}(x), y) \leq \ell_{\max},$$

and that for each class y , the data distribution is supported on a semantic manifold $\mathcal{M}_y$ in feature space, while regions far from $\mathcal{M}_y$ carry negligible probability mass.

Proof. We formalize the intuition described in the main text. Under long-tailed distributions, adversarial optimization may place excessive emphasis on perturbations whose features drift far away from the semantic manifold $\mathcal{M}_y$ of a tail class. Although such off-manifold perturbations can induce high classification loss, they lie in low-density regions that are weakly supported by the data distribution and therefore do not meaningfully contribute to robustness on the semantic support.

Fix a sample (x, y) and consider the inner maximization in the robust risk. Let

$$x^* = \arg \max_{x' \in B_\epsilon(x)} \ell(f_\Theta(x'), y).$$

Step 1: Decomposition of robust risk. To make the above distinction precise, we decompose the robust risk into on-manifold and off-manifold contributions:

$$R_{\text{robust}}(\Theta) = R_{\text{on}}(\Theta) + R_{\text{off}}(\Theta),$$

where R_{on} corresponds to adversarial examples whose features remain within a neighborhood of the class manifold $\mathcal{M}_y$, and R_{off} corresponds to adversarial examples whose features drift far away from $\mathcal{M}_y$.

By the manifold support assumption, off-manifold regions contribute negligible probability mass. Therefore, there exists a constant $\rho \ll 1$ such that

$$R_{\text{off}}(\Theta) \leq \rho \ell_{\max}.$$

Step 2: Control of on-manifold risk via manifold-constrained objective. Recall the MCAT objective without the geometric regularizer:

$$R_{\text{MCAT}}(\Theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta\|_\infty \leq \epsilon} (\ell(f_\Theta(x + \delta), y) + \lambda d_{\mathcal{M}_y}(\phi_\Theta(x + \delta))) \right].$$

For adversarial examples whose features lie within a bounded neighborhood of $\mathcal{M}_y$, the manifold deviation term is uniformly bounded:

$$d_{\mathcal{M}_y}(\phi_\Theta(x')) \leq C.$$

As a result, for such on-manifold adversarial points,

$$\begin{aligned} \ell(f_\Theta(x'), y) &\leq \ell(f_\Theta(x'), y) + \lambda d_{\mathcal{M}_y}(\phi_\Theta(x')) \\ &\leq R_{\text{MCAT}}(\Theta) + \frac{C}{\lambda}. \end{aligned}$$

Step 3: Combining the bounds. Taking expectation over (x, y) and combining the on-manifold and off-manifold contributions yields

$$R_{\text{robust}}(\Theta) \leq R_{\text{MCAT}}(\Theta) + \frac{C}{\lambda} + \rho \ell_{\max}.$$

Since ρ is negligible under the manifold support assumption, we conclude that

$$R_{\text{robust}}(\Theta) \leq R_{\text{MCAT}}(\Theta) + O(\lambda^{-1}),$$

which completes the proof. $\square$

B Generator Architecture

Each class-conditional generator G_y is implemented as a lightweight multilayer perceptron operating in the classifier feature space. Given a latent code $z \in \mathbb{R}^{d_z}$ sampled from a standard Gaussian, G_y outputs a feature vector in $\mathbb{R}^{d_f}$, where d_f is the dimension of the penultimate-layer features of the backbone.

Concretely, G_y consists of three fully connected layers with widths $d_z \rightarrow 1024 \rightarrow 1024 \rightarrow d_f$. ReLU activations are applied after the first two layers, and the output layer is linear. No batch normalization or dropout is used. All generators share the same architecture but are trained independently for each class. Unless otherwise specified, we set $d_z = 128$ and $d_f = 512$ for CIFAR-based experiments.

C More Experimental Results

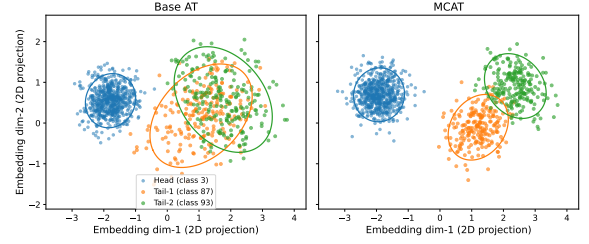


Figure 8: Case study on CIFAR-100-LT (IR=100) showing 2D embedding projections of one head and two tail classes. Compared to Base AT, MCAT yields tighter and better-separated tail clusters while maintaining compact head representations.

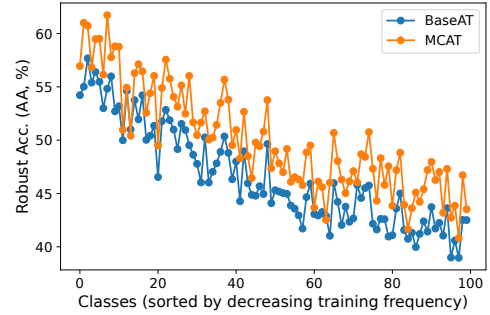
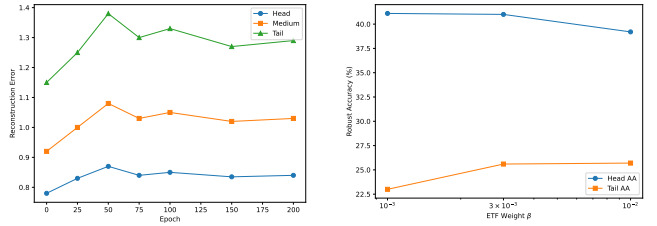


Figure 9: Robust accuracy over all classes under AutoAttack (AA) on CIFAR-100-LT (IR=100), plotted against the sorted class frequency rank.



(a) Reconstruction error $\|\phi_\Theta(x) - G_y(z^*)\|_2$ over training epochs on CIFAR-100-LT (IR=100). The error exhibits mild non-monotonic fluctuations due to feature adaptation, but remains stable overall and does not diverge for tail classes.

(b) Effect of ETF regularization weight β on robust accuracy for head and tail classes. Moderate β improves tail robustness without degrading head performance, while excessively large β leads to over-regularization.

Figure 10: Manifold stability and geometric alignment trade-off on CIFAR-100-LT (IR=100). Left: reconstruction error of frozen class-conditional generators remains stable throughout adversarial training. Right: ETF-inspired geometric alignment improves tail robustness without sacrificing head-class performance under moderate regularization.

Method	IR=10	IR=20	IR=50	IR=100
RoBal	31.80	29.10	26.40	25.90
Self-Distill	33.40	31.20	29.10	28.10
AT-BSL	32.90	30.80	28.70	27.30
MCAT (ours)	34.60	33.10	32.00	31.50

Table 4: AutoAttack robustness (%) on CIFAR-100-LT under varying imbalance ratios. MCAT exhibits consistently higher robustness and degrades more gracefully as imbalance severity increases.

Method	IR=10	IR=20	IR=50	IR=100
RoBal	19.60	17.40	14.80	13.20
Self-Distill	21.30	19.50	17.40	16.10
AT-BSL	20.80	18.90	16.90	15.50
MCAT (ours)	26.20	23.60	22.10	21.05

Table 5: Tail-class AutoAttack robustness (%) on CIFAR-100-LT under varying imbalance ratios. MCAT consistently improves tail robustness and degrades more gracefully as imbalance severity increases.

T_z	Overall AA (%)	Tail AA (%)	Relative Train Time
0	32.4	18.7	1.00
1	34.9	21.5	1.07
3	36.8	24.3	1.14
5	37.5	25.6	1.18
8	37.6	25.7	1.26

Table 6: Sensitivity to the number of latent optimization steps T_z on CIFAR-100-LT (IR=100).

Method	BA $\uparrow$	BR (AA) $\uparrow$	Tail-PGD $\uparrow$	Tail-AA $\uparrow$
PGD-AT	31.20 $\pm$ 0.60	11.40 $\pm$ 0.65	9.30 $\pm$ 0.70	8.10 $\pm$ 0.75
TRADES	32.10 $\pm$ 0.58	12.80 $\pm$ 0.62	10.20 $\pm$ 0.68	8.90 $\pm$ 0.73
MART	31.80 $\pm$ 0.59	13.20 $\pm$ 0.63	10.60 $\pm$ 0.69	9.20 $\pm$ 0.74
AWP	32.90 $\pm$ 0.55	14.10 $\pm$ 0.60	11.30 $\pm$ 0.65	10.10 $\pm$ 0.70
RoBal	35.60 $\pm$ 0.52	15.90 $\pm$ 0.58	12.80 $\pm$ 0.62	11.20 $\pm$ 0.67
REAT	36.80 $\pm$ 0.50	17.10 $\pm$ 0.56	13.90 $\pm$ 0.60	12.30 $\pm$ 0.65
TAET	37.40 $\pm$ 0.48	17.80 $\pm$ 0.55	14.60 $\pm$ 0.59	12.90 $\pm$ 0.64
Self-Distill	37.90 $\pm$ 0.49	18.20 $\pm$ 0.54	15.00 $\pm$ 0.58	13.40 $\pm$ 0.63
AT-BSL	37.60 $\pm$ 0.50	17.90 $\pm$ 0.55	14.70 $\pm$ 0.59	13.10 $\pm$ 0.64
MCAT (ours)	42.30$\pm$0.45	22.60$\pm$0.50	18.90$\pm$0.54	16.80$\pm$0.58

Table 7: Balanced and tail robustness on **Tiny-ImageNet-LT (IR=100)**. We report balanced accuracy (BA) and balanced robustness (BR), defined as average per-class accuracy under clean evaluation and AutoAttack (AA), respectively, together with tail-class robust accuracy under PGD-20 (Tail-PGD) and AutoAttack (Tail-AA). Results are reported as mean $\pm$ std over three random seeds.

D Hyperparameter Settings

Category	Hyperparameter	Value
Training	Optimizer	SGD with momentum
	Momentum	0.9
	Weight decay	5×10^{-4}
	Batch size	128
	Training epochs	200
Adversarial Setup	Learning rate schedule	Cosine decay
	Threat model	ℓ_∞
	Perturbation budget ϵ	8/255
	Step size α	2/255
Backbone	PGD steps	10 (train), 20 (eval)
	Architecture	ResNet-18
	Initialization	He initialization
Long-tailed Setting	Normalization	BatchNorm
	Imbalance type	Exponential
	Imbalance ratio (IR)	{10, 50, 100}
	Sampling strategy	Class-uniform
MCAT (Ours)	Evaluation metric	Overall / Many / Medium / Few
	Manifold penalty weight λ_{man}	0.1
	Geometric alignment weight β	3×10^{-3}
	Equivalent λ_{geom} in implementation	0.01
	Manifold distance metric	ℓ_2 in feature space
	Manifold update frequency	Every iteration
	ETF target dimension	Equal to number of classes
Baselines	AT / TRADES	Official recommended settings
	RoBal	Margin reweighting as in [Wu <i>et al.</i> , 2021]
	REAT	Loss reweighting as in [Li <i>et al.</i> , 2023]
	TAET	Two-stage schedule as in [Yu-Hang <i>et al.</i> , 2025]
	Distillation-based	Teacher trained on balanced AT

Table 8: Hyperparameter settings for MCAT and baseline adversarial training methods. β denotes the ETF-inspired geometric alignment weight used in Eq. (3). Unless otherwise specified, all methods share the same backbone architecture and training protocol.